

A From FIFO to the Recovered Front

Figure 1 makes the central allocation result legible at a glance. The incumbent first-in, first-out (FIFO) policy and the 30-run combined Pareto front are placed in the full three-objective space *(a)* and then read one objective at a time *(b)*. FIFO is Pareto-dominated: it wastes 1,940 visas, whereas 94 of the 104 front solutions waste none; its maximum inter-country disparity of 13.0 years can be cut to 2.0 years at a cost of only $\sim 2.4\%$ in the unserved waiting load f_1 , on which FIFO is already near-optimal and the front merely ties it. No single solution attains all three extremes at once—that is precisely the trade-off the front quantifies.

A striking corollary concerns how *little* reallocation domination requires (Figure 2). The min- f_1 policy—the one that Pareto-dominates FIFO—rescues 1,260 of the incumbent’s 1,940 wasted visas and puts them to use (reaching 139,320 of the 140,000 visas), and it does so by changing only 2 of the 21 country totals; the remaining 19 are unchanged and no objective is worse than FIFO. Equity-oriented points on the front (e.g., the min- f_2 extreme) redistribute far more—the structural tension the front quantifies—so we display the dominating, minimal-disruption policy here. All values are under the assumed synthetic model and constitute auditable decision support; the totals illustrate method behavior, never a recommended allocation.

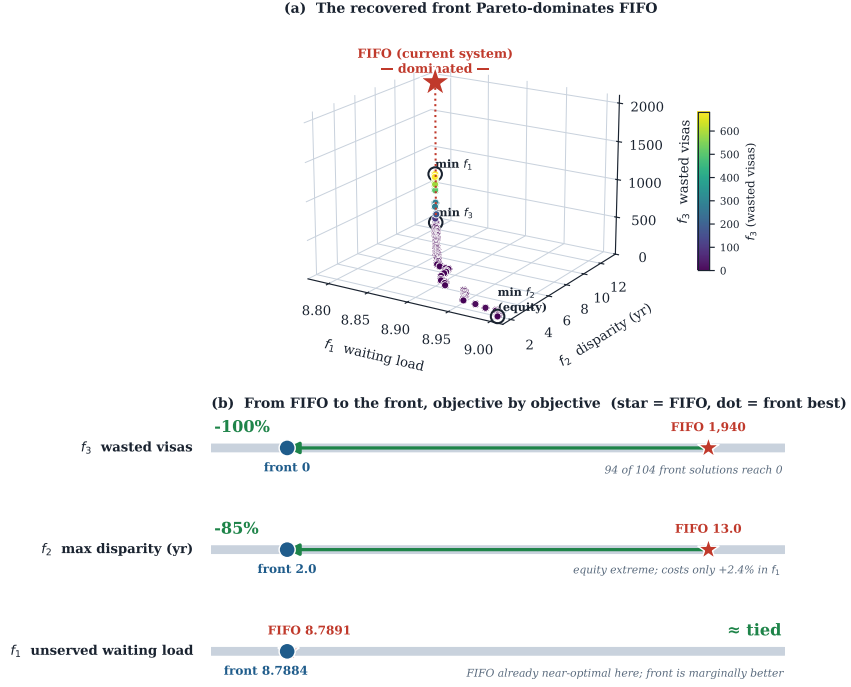


Fig. 1. From the FIFO incumbent to the recovered front (seeds 1–30). (a) The 104-solution combined front in the full objective space, colored by wasted visas f_3 ; the FIFO baseline (star) floats above the front and is Pareto-dominated, with the three extreme solutions ringed. (b) The same comparison objective by objective: wasted visas fall from 1,940 to 0 (94/104 solutions), maximum disparity from 13.0 to 2.0 years (the equity extreme, costing only +2.4% in f_1), while f_1 essentially ties FIFO (the front is marginally better).

The min- f_1 policy Pareto-dominates FIFO: more visas used, minimal footprint

+1,260 visas rescued from waste and put to use

139,320 of 140,000 visas used (FIFO uses 138,060)

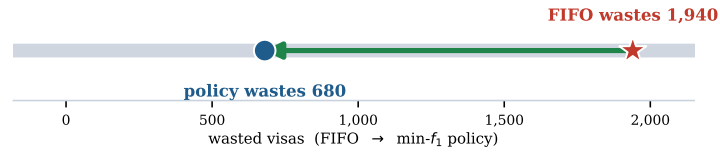


Fig. 2. How the min- f_1 policy dominates FIFO (seeds 1–30, firewall-verified front). It rescues 1,260 of FIFO’s 1,940 wasted visas (reaching 139,320 of 140,000 used) and changes only 2 of the 21 country totals, with no objective worse than FIFO. The totals illustrate method behavior under the assumed synthetic model, not a recommended allocation.